

Here is the **fully reorganized and consolidated Bibliography Checklist** for our PetNet paper.

This list integrates standard academic references with the specific engineering resources (COCO, CC0 APIs) found in our codebase. I have categorized them logically to match the flow of a top-tier conference paper (Introduction → Related Work → Methodology → Experiments).

1. The Problem Context: Fine-Grained Visual Categorization (FGVC)

Use these in the **Introduction** to explain why classifying pet breeds is difficult and why standard global feature extraction fails.

- **[1] Bilinear CNN Models for Fine-grained Visual Recognition**
 - *Lin, T. Y., RoyChowdhury, A., & Maji, S. (ICCV 2015)*
 - **Key Theory:** Uses second-order statistics to model part-feature interactions.
 - **Role in Project:** Cited as the pioneering work in FGVC. we argue that while effective, Bilinear CNNs are computationally too heavy for mobile devices, justifying our lightweight **PetNet**.
 - **Paper Link:** [CVF PDF](#)
 - **[2] Learning to Navigate for Fine-grained Classification (NTS-Net)**
 - *Yang, Z., Luo, T., Wang, D., Hu, Z., Gao, J., & Wang, L. (ECCV 2018)*
 - **Key Theory:** Uses a "Navigator" agent to identify informative regions (like ears/eyes) and a "Teacher" to classify them.
 - **Role in Project:** Represents the "Multi-Stage" approach. we contrast this with **LDRE**: "Instead of using a heavy region-proposal network like NTS-Net, we use efficient pre-computed masks (LDRE) to achieve similar focus on detailed parts."
 - **Paper Link:** [arXiv PDF](#)
 - **[3] Destruction and Construction Learning for Fine-grained Image Recognition (DCL)**
 - *Chen, Y., Bai, Y., Zhang, W., & Mei, T. (CVPR 2019)*
 - **Key Theory:** Jigsaw-puzzle-style destruction forces the model to learn semantic relationships between parts.
 - **Role in Project: Theoretical Support for LDRE.** we cite this to support the hypothesis that "destroying local structures (masking) forces the network to learn global robust features."
 - **Paper Link:** [CVF PDF](#)
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2. The Foundation: Lightweight Architectures

Use these in **Methodology** to describe our backbone choices and comparisons.

- **[4] MobileNetV2: Inverted Residuals and Linear Bottlenecks**
 - Sandler, M., Howard, A., Zhu, M., Zhmoginov, A., & Chen, L. C. (CVPR 2018)
 - **Key Theory:** Inverted Residual Blocks (IRB) and Depthwise Separable Convolutions to reduce FLOPs.
 - **Role in Project: The Backbone.** This is the foundation of PetNet. we explicitly state that we modify the standard MBV2 by adding attention mechanisms and distillation heads.
 - **Paper Link:** [CVF PDF](#)
 - **[5] EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks**
 - Tan, M., & Le, Q. (ICML 2019)
 - **Key Theory:** Compound scaling of depth, width, and resolution.
 - **Role in Project: Comparison Target.** Used to represent "Heavyweight SOTA." we show that PetNet achieves comparable accuracy to larger models with a fraction of the parameters.
 - **Paper Link:** [arXiv PDF](#)
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3. Methodology A: Data Augmentation (Occlusion & Mixing)

Use these in **Related Work** and **Methodology (LDRE)** to highlight our innovation.

- **[6] Random Erasing Data Augmentation**
 - Zhong, Z., Zheng, L., Kang, G., Li, S., & Yang, Y. (AAAI 2020)
 - **Key Theory:** Randomly replacing rectangular regions with noise improves robustness.
 - **Role in Project: The "Straw Man" (Comparison Target).** we argue that Random Erasing is sub-optimal because it blindly erases background or useful features. LDRE is superior because it is *semantically guided*.
 - **Paper Link:** [arXiv PDF](#)
- **[7] GridMask Data Augmentation**
 - Chen, P., Liu, S., Zhao, H., & Jia, J. (arXiv 2020)
 - **Key Theory:** Using a structured grid for occlusion avoids masking the entire object.
 - **Role in Project:** Precursor to LDRE. we adopt the "grid" concept but refine it with "keypoint probability" (Anatomical Prior).
 - **Paper Link:** [arXiv PDF](#)
- **[8] mixup: Beyond Empirical Risk Minimization**
 - Zhang, H., Cisse, M., Dauphin, Y. N., & Lopez-Paz, D. (ICLR 2018)

- **Key Theory:** Linear interpolation of images and labels ($x' = \lambda x_i + (1 - \lambda)x_j$) smooths the decision boundary.
 - **Role in Project: Synergistic Partner.** we explain that MixUp handles pixel-distribution shift, while LDRE handles structural overfitting. Together, they achieve SOTA.
 - **Paper Link:** [arXiv PDF](#)
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4. Methodology B: Architecture Modules (Attention & Distillation)

Use these to explain the internal components of PetNet.

- **[9] ECA-Net: Efficient Channel Attention for Deep Convolutional Neural Networks**
 - Wang, Q., Wu, B., Zhu, P., Li, P., Zuo, W., & Hu, Q. (CVPR 2020)
 - **Key Theory:** 1D convolution to determine channel importance without dimensionality reduction.
 - **Role in Project:** The source of our **Dual-Attention** module. we chose it over SE-Net for its low parameter cost.
 - **Paper Link:** [arXiv PDF](#)
 - **[10] Be Your Own Teacher: Improve the Performance of Convolutional Neural Networks via Self Distillation**
 - Zhang, L., Song, J., Gao, A., Chen, J., Bao, C., & Ma, K. (ICCV 2019)
 - **Key Theory:** Using the deepest layer of the network to supervise shallower layers (Auxiliary Heads).
 - **Role in Project: Theoretical Basis for SelfKD.** we cite this to explain why we added auxiliary loss heads to Stage 1 and Stage 2 of PetNet.
 - **Paper Link:** [arXiv PDF](#)
 - **[11] Attention Is All You Need**
 - Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., ... & Polosukhin, I. (NIPS 2017)
 - **Key Theory:** Self-Attention and Positional Encodings.
 - **Role in Project:** Cited to explain the **2D Positional Encoding** component within our Dual-Attention module (helping the CNN understand spatial geometry).
 - **Paper Link:** [arXiv PDF](#)
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5. Implementation: Domain-Specific Pose Datasets

*Use these in **Experiments** to explain how we trained the Keypoint Detector (YOLOv8-Pose) for LDRE.*

- **[12] Who Left the Dogs Out? 3D Animal Reconstruction (StanfordExtra)**
 - *Biggs, B., Boyne, O., Charles, J., Fitzgibbon, A., & Cipolla, R. (ECCV 2020)*
 - **Role in Project: Crucial for LDRE.** Standard human-pose models fail on dogs. we cite this to explain how we fine-tuned YOLOv8 to accurately detect dog ears/noses for our masking algorithm.
 - **Paper Link:** [arXiv PDF](#)
 - **Project:** [GitHub](#)
 - **[13] Cross-Domain Adaptation for Animal Pose Estimation (Animal Pose Dataset)**
 - *Cao, J., Tang, H., Fang, H. S., Shen, X., Lu, C., & Tai, Y. W. (ICCV 2019)*
 - **Role in Project: Crucial for LDRE.** Used to ensure our keypoint detector works for **Cats** as well as **Dogs**, ensuring the LDRE module is generic for all pets.
 - **Paper Link:** [arXiv PDF](#)
 - **Project:** [GitHub](#)
 - **[14] AP-10K: A Benchmark for Animal Pose Estimation in the Wild**
 - *Yu, H., Xu, Y., Zhang, J., Zhao, W., Guan, Z., & Tao, D. (NeurIPS 2021)*
 - **Role in Project:** General reference for animal pose estimation standards and pre-training data.
 - **Paper Link:** [arXiv PDF](#)
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6. Classification Benchmarks & Supplementary Data

Use these in *Experiments* and *Data Preparation*.

- **[15] Cats and Dogs (Oxford-IIIT Pet Dataset)**
 - *Parkhi, O. M., Vedaldi, A., Zisserman, A., & Jawahar, C. V. (CVPR 2012)*
 - **Role in Project:** Primary training/testing dataset.
 - **Paper Link:** [Official PDF](#)
- **[16] Novel Dataset for Fine-Grained Image Categorization (Stanford Dogs)**
 - *Khosla, A., Jayadevaprakash, N., Yao, B., & Fei-Fei, L. (CVPR 2011 Workshop)*
 - **Role in Project:** Primary fine-grained classification dataset.
 - **Paper Link:** [Official PDF](#)
- **[17] Microsoft COCO: Common Objects in Context**
 - *Lin, T. Y., et al. (ECCV 2014)*
 - **Role in Project: Foundational Data.** Used in `data_downloader.py` to fetch diverse background images or for pre-training the detector to ensure robustness against non-pet objects.
 - **Paper Link:** [arXiv PDF](#)
- **[18] Public Domain Image Sources (Unsplash, Pexels, Pixabay)**
 - **Role in Project: Robustness Testing.** We utilized custom scripts (`scarpe_cc0_api.py`) to harvest high-resolution, complex-scene images from

these platforms. This allowed us to create a "Wild" test set to validate PetNet's performance in unconstrained environments.

- **URLs:** [Unsplash API](#), [Pexels API](#), [Pixabay API](#)
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7. Tools & Visualization

- **[19] Ultralytics YOLO**
 - *Jocher, G., et al.*
 - **Role in Project:** The engine used for object detection and keypoint extraction.
 - **Project:** [GitHub](#)
- **[20] Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization**
 - *Selvaraju, R. R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., & Batra, D. (ICCV 2017)*
 - **Role in Project: Qualitative Analysis.** The method used to generate heatmaps (via `visualize_gradcam.py`) to visually prove that LDRE forces the model to look at the whole body.
 - **Paper Link:** [arXiv PDF](#)